

# A Brief Introduction to Data Communication and Networking

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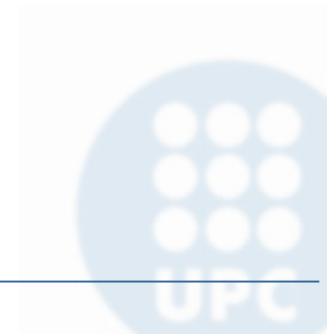
Mónica Aguilar Igartua

*[monica.aguilar@upc.edu](mailto:monica.aguilar@upc.edu)*

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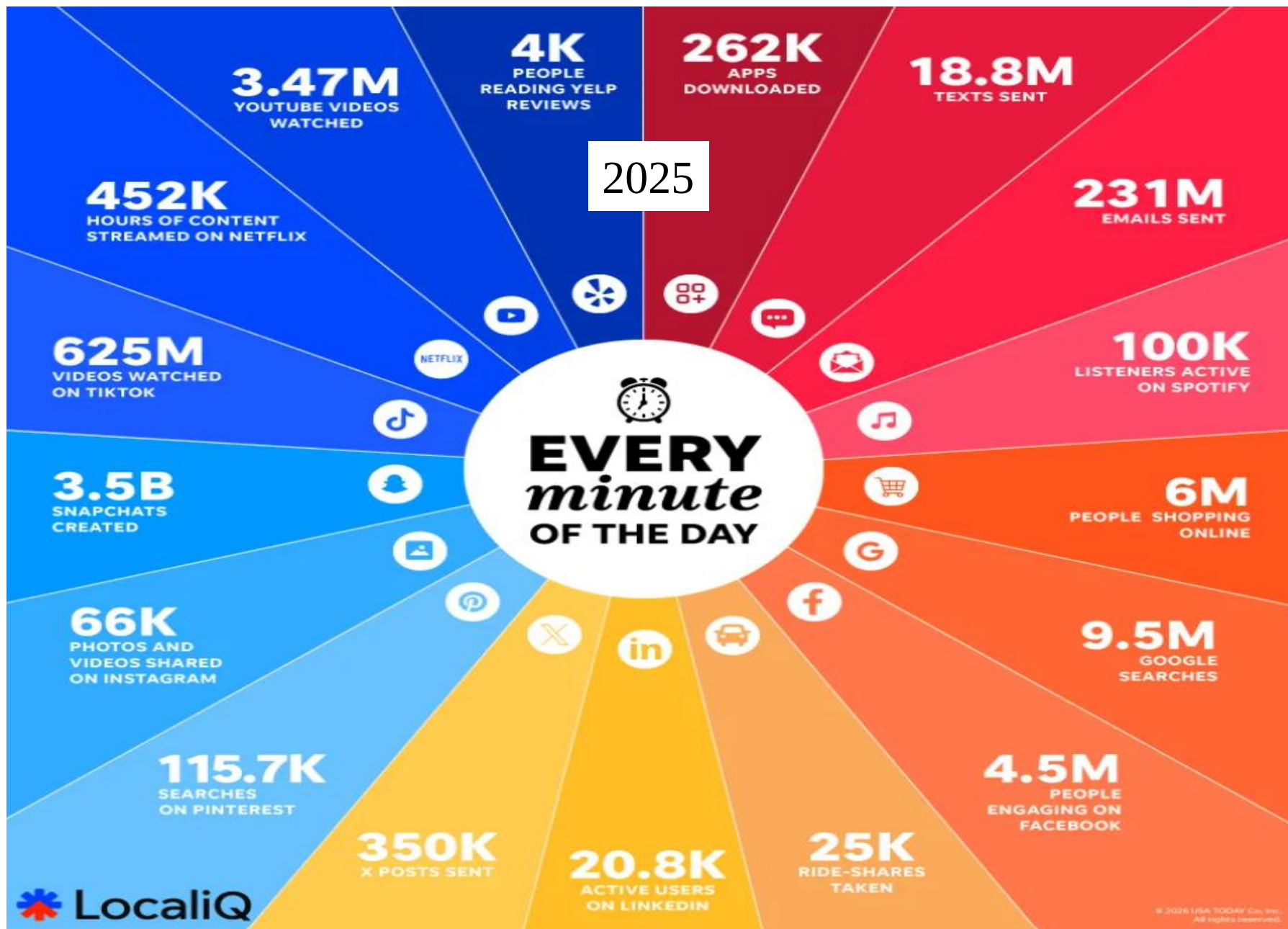
# Contents

- How can we transmit information?
- Components of a communication
- Classify the tasks: Layering model



# Com funciona Internet?





# Per a què ens serveix Internet?



# Com funciona Internet?



**Receiver**

**Sender**

Transmission

Medium

**Message**



# Components d'una comunicació

Without a protocol we could not understand the message.

We need to agree:

When do we receive

Language

Receiver

Transmission  
Medium

Sender

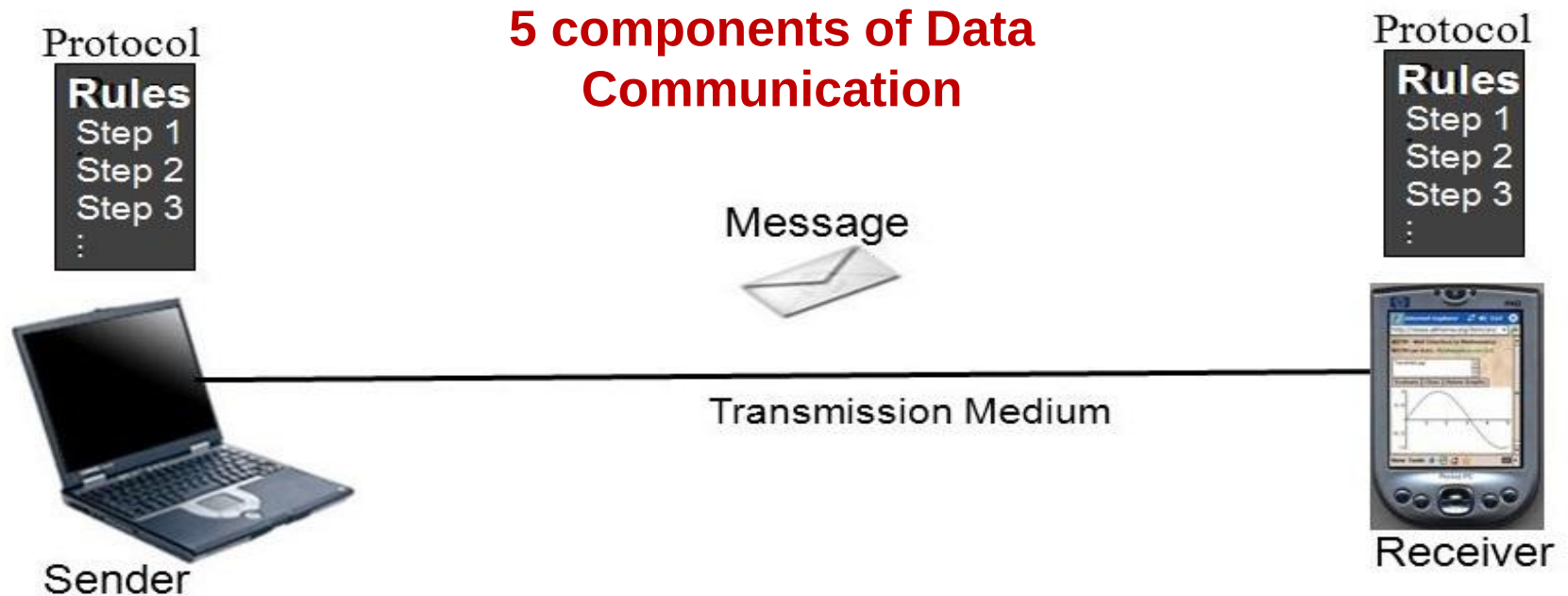
Message

When do we send



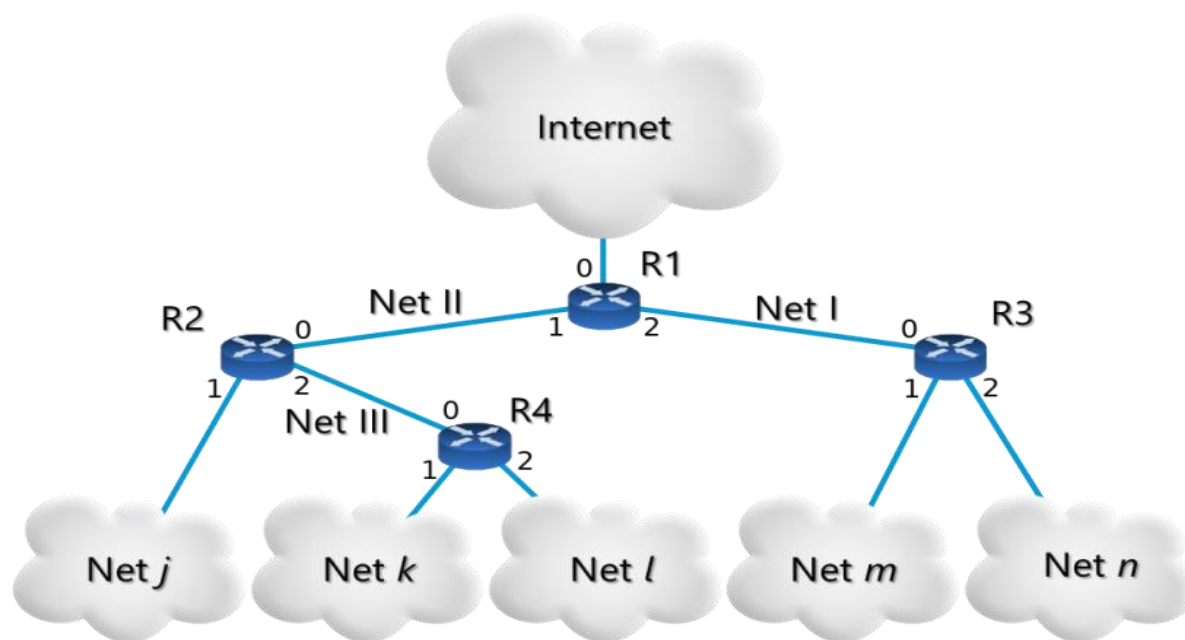
**Protocol = a set of rules, an agreement**

# Com funciona Internet?



- We need to define the **devices** (hardware), the **medium** and the instructions (**protocol**), which allow us to transmit and process the **information** correctly.
- Without a protocol, two devices may be connected, but will not be able to communicate each other.

# Networks and Internetworking

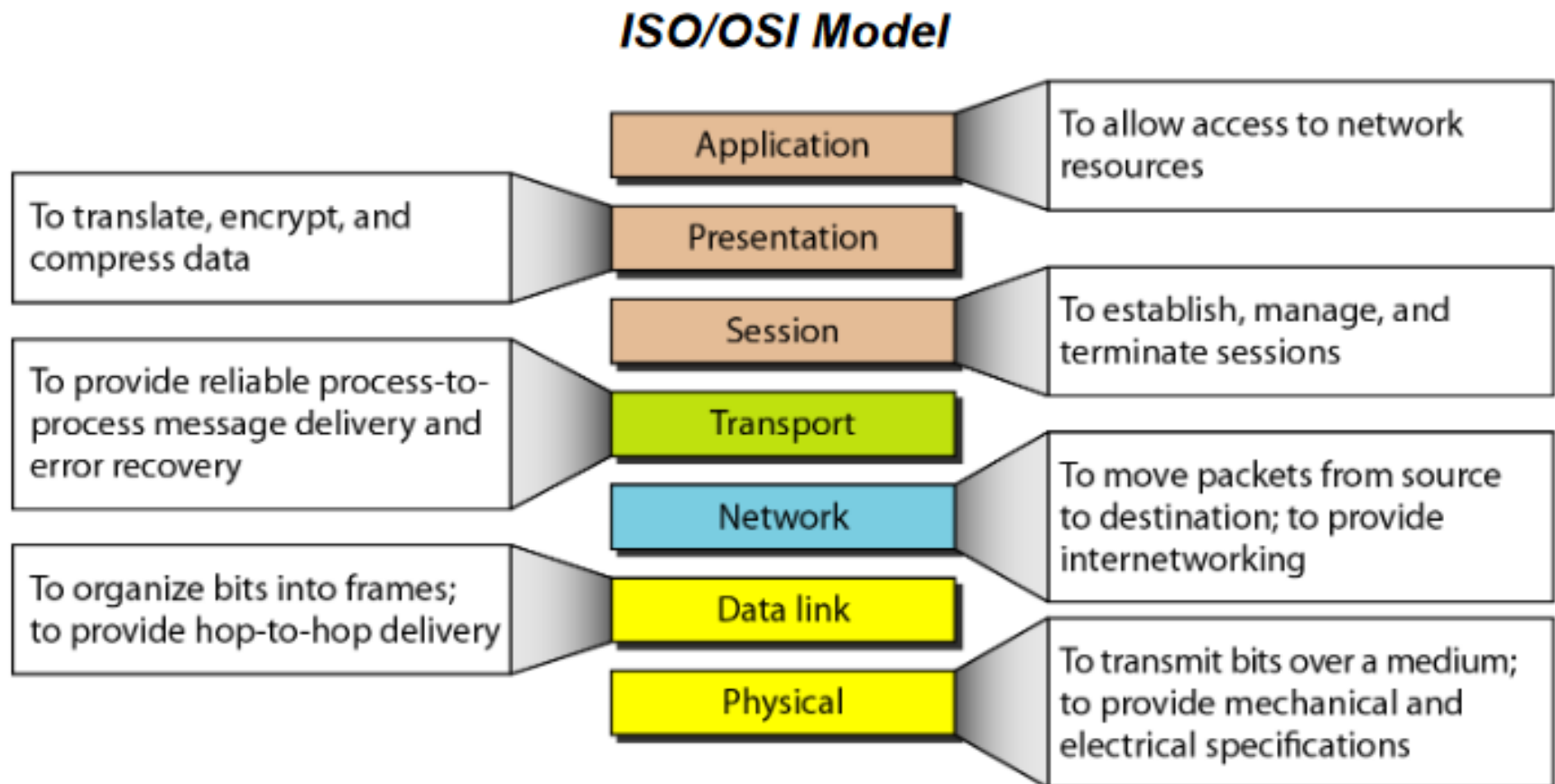


- Interconnection of devices
  - Resources are limited.
    - Share resources (printers, servers, routers...)
  - Communication over longer distances
    - Devices such as **routers** and **gateways**



# Layering Model

- Each level has its own goals
- Flexible to update and improve at each level independently

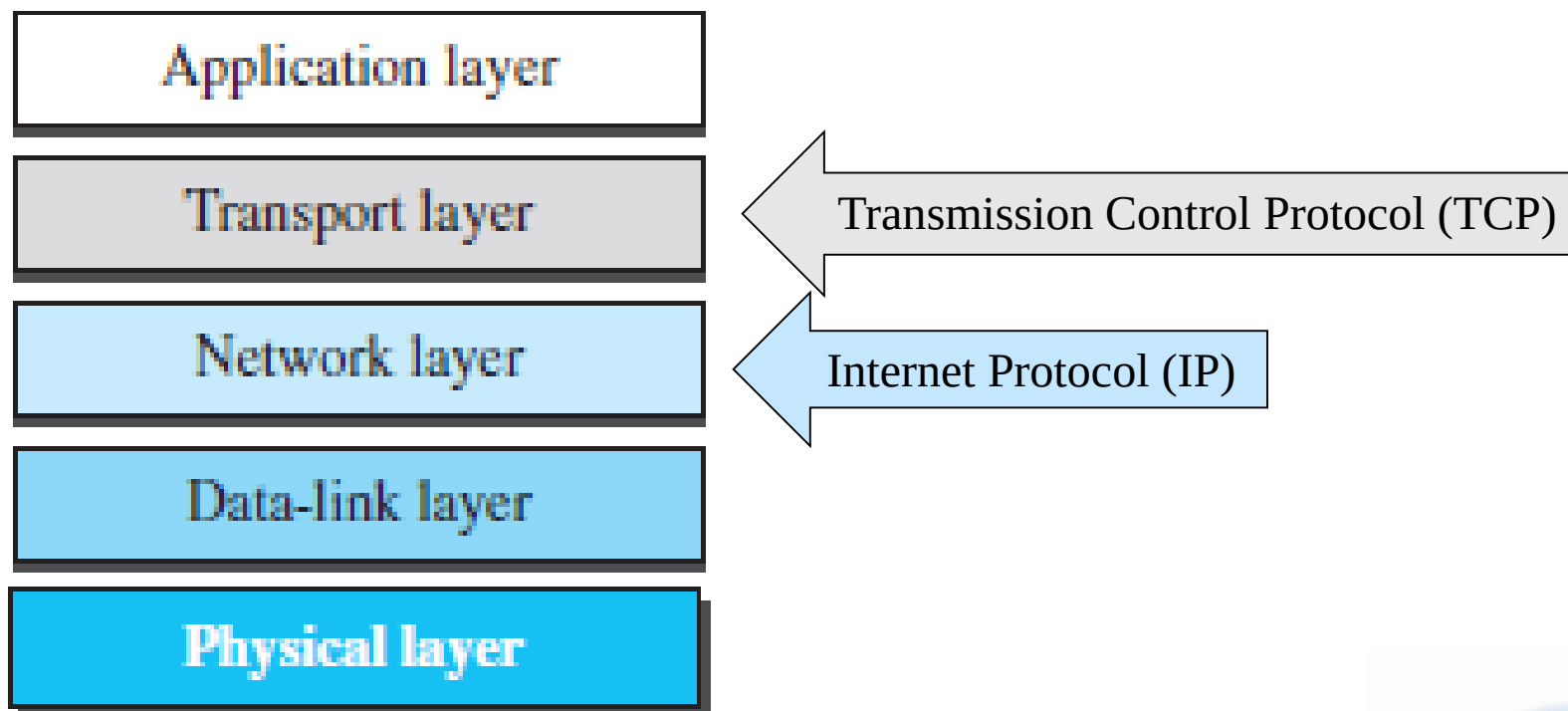


Fuente: "Data Communications and Networking", B. A. Fourozan, 4th edition, capítulo 2..

# The Protocol Layering of the Internet

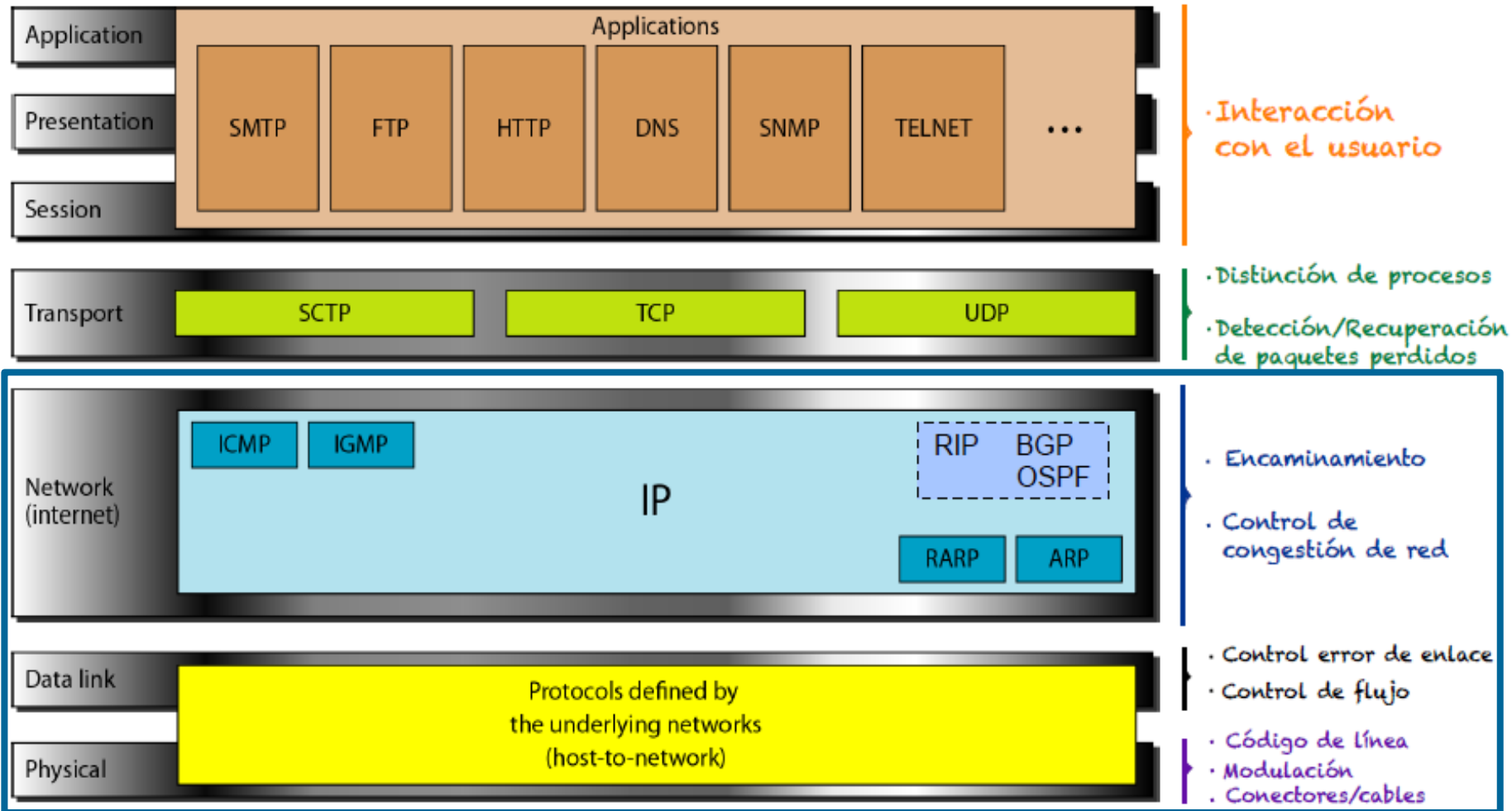
## TCP/IP protocol stack

### Layers



To define the tasks, we need **protocols** at each layer

# Network Models: TCP/IP and OSI model



IP: Internet Protocol

SCTP: Stream Control Transmission Protocol

TCP: Transmission Control Protocol

UDP: User Datagram Protocol